OSI Layer 2: Data Link Layer

1. Introduction to the Data Link Layer

The Data Link Layer (Layer 2) is the second layer of the OSI reference model. It defines how data is organized into frames and exchanged between devices connected to the same local network link.

While the Physical Layer (Layer 1) is responsible for transmitting and receiving raw bits over a physical medium, the Data Link Layer provides the mechanisms needed to transfer data between neighboring network interfaces.

A simple way to understand the difference is:

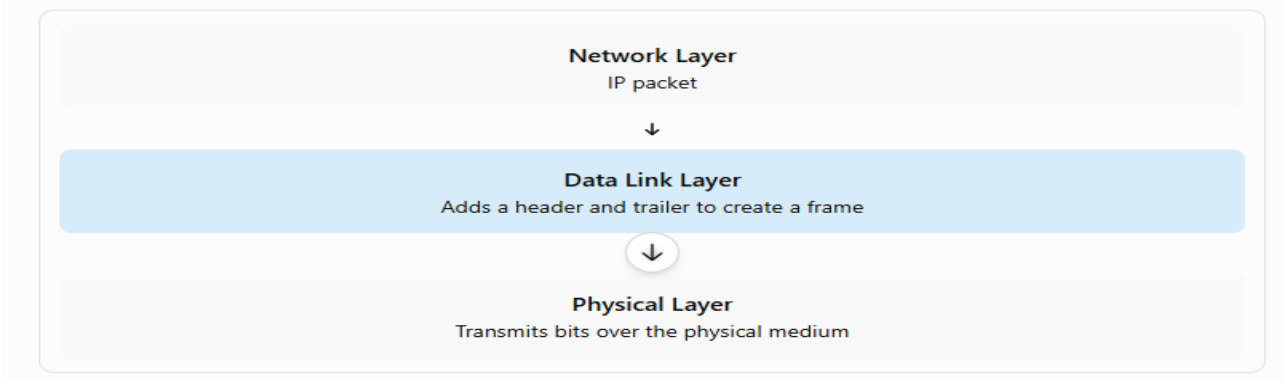
- Physical Layer: Transmits and receives bits.
- Data Link Layer: Organizes bits into frames and manages their delivery across a local link.

For example, when a computer sends an IP packet through an Ethernet network, the Data Link Layer encapsulates the packet inside an Ethernet frame. The frame contains information such as the source MAC address, destination MAC address, and an error-detection field.

1.1 Data Link Layer Communication

The Data Link Layer supports communication between directly connected network interfaces. These interfaces may belong to computers, routers, switches, or other network devices.

The basic process is:



At the receiving device, the reverse process occurs. The Physical Layer receives the bits, and the Data Link Layer reconstructs and processes the frame. If the frame is valid and intended for the receiving interface, its payload can be passed to the appropriate upper-layer protocol.

Important terminology: The Data Link Layer Protocol Data Unit (PDU) is called a frame. The Network Layer PDU is called a packet, and the Physical Layer transmits bits.

2. Main Functions of the Data Link Layer

Data Link Layer protocols may provide several functions, depending on the technology and protocol being used.

1. Framing
2. Media Access Control (MAC)
3. Addressing
4. Error Detection
5. Identification of Encapsulated Data
6. Flow and Error Control

2.1 Framing

Framing is the process of organizing data received from an upper layer (Network layer) into a structured unit called a frame. A frame normally includes:

- A header containing control information, such as source and destination addresses.
- A payload containing the encapsulated data.
- A trailer or other error-detection information, depending on the protocol.

Framing allows the receiver to determine where a data unit begins and ends and how it should be processed.

2.2 Media Access Control (MAC)

- When multiple devices share a transmission medium, they need rules that determine when each device may transmit. These rules are called media access control mechanisms.
- For example, traditional shared Ethernet used CSMA/CD, while Wi-Fi uses CSMA/CA.
- In modern full-duplex switched Ethernet, each device typically has a dedicated link to a switch, so devices do not compete with one another for the same collision-prone medium.

2.3 Addressing

The Data Link Layer uses physical addresses, commonly called MAC addresses in Ethernet, to identify network interfaces on a local network.

The destination MAC address helps a receiving device or switch determine whether a frame should be accepted or forwarded.

2.4 Error Detection

- Transmission may introduce bit errors because of noise, interference, or other physical problems.

- Many Data Link Layer protocols use an error-detection mechanism to determine whether a received frame has been corrupted.
- Ethernet, for example, uses a Frame Check Sequence (FCS) based on a Cyclic Redundancy Check (CRC).
- Error detection does not necessarily mean that the Data Link Layer retransmits the frame.

2.5 Identification of Encapsulated Data

- A frame may carry data belonging to different upper-layer protocols.
- The receiver needs a way to determine which protocol should process the payload.
- For example, Ethernet II uses an Ether Type field to identify protocols such as IPv4, ARP, and IPv6.

2.6 Flow and Error Control

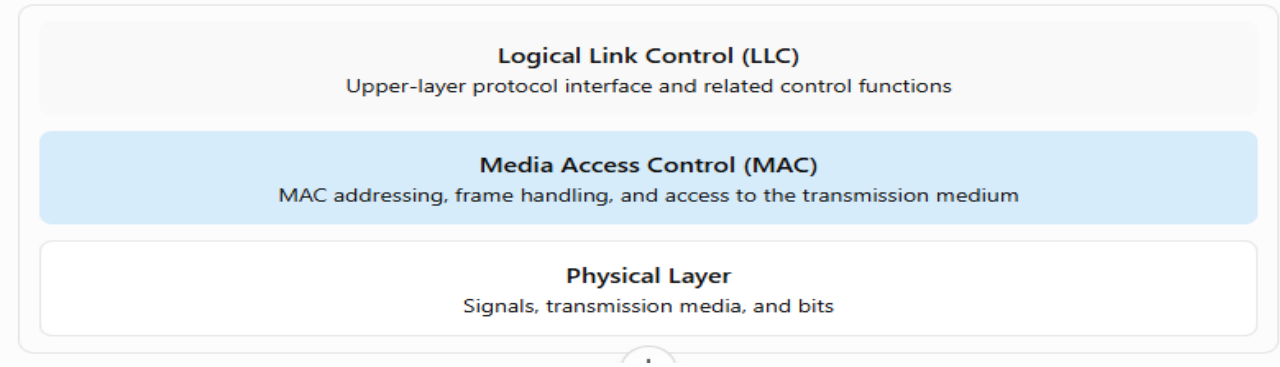
Some Data Link Layer protocols provide flow control, error recovery, or both. These services are protocol-dependent and are not provided in the same way by every Layer 2 technology.

For example, Ethernet's FCS detects corrupted frames, but ordinary Ethernet does not provide link-layer retransmission of corrupted frames in the same way as certain other link protocols.

3. Data Link Sublayers: MAC and LLC

In the IEEE 802 architecture, the Data Link Layer is commonly described as having two sublayers.

1. Logical Link Control (LLC)
2. Media Access Control (MAC)



3.1 Media Access Control (MAC) Sublayer

The MAC sublayer deals with access to the transmission medium and link-layer addressing.

Its responsibilities commonly include:

- Defining MAC addressing and its use in frames.
- Managing access to a shared medium where required.
- Supporting frame transmission and reception.
- Providing error-detection information in technologies such as Ethernet.

The exact responsibilities depend on the Data Link Layer technology.

3.2 Logical Link Control (LLC) Sublayer

- The LLC sublayer provides a logical interface between certain IEEE 802 Data Link Layer services and the Network Layer.
- It can help identify upper-layer protocols and support link-control services defined by the relevant protocol.
- Technical note: Do not teach that all encapsulated-data identification is performed by LLC. Ethernet II frames use the EtherType field directly to identify the encapsulated protocol.

4. MAC Methods and Media Sharing

The method used to control access to a medium depends mainly on two factors.

- **Media sharing**: Whether several devices use the same medium and must coordinate their transmissions.
- **Topology**: How devices are connected and how that connection appears to the communication protocol.

When several devices transmit over the same shared medium without coordination, their signals may interfere. This interference can cause a collision in collision-prone technologies.

In computer networks, MAC (Media Access Control) methods determine how multiple devices share and access a common transmission medium, such as a communication channel.

The two main categories are:

- Controlled Access
- Contention-Based Access

Controlled Access

In controlled access, devices coordinate with one another to determine which device is allowed to transmit at a particular time. This helps prevent collisions.

Main methods:

- Reservation: Devices reserve time slots before transmitting.

- Polling: A central controller asks each device in turn whether it has data to send.
- Token Passing: A special control frame called a token circulates among devices. Only the device holding the token can transmit.

Example: In token-passing networks, a device must receive the token before sending data.

Contention-Based Access

In contention-based access, devices compete for access to the shared medium. A device attempts to transmit when it determines that the medium is available. Collisions may occur if two or more devices are transmitted simultaneously. Main methods:

- ALOHA: A device transmits when it has data to send; collisions are handled through retransmission.
- CSMA (Carrier Sense Multiple Access): A device listens to the medium before transmitting.
- CSMA/CD (Collision Detection): Detects collisions and manages retransmission. Historically used in shared, half-duplex Ethernet.
- CSMA/CA (Collision Avoidance): Attempts to avoid collisions, particularly in Wi-Fi networks.

Feature	Controlled Access	Contention-Based Access
Access mechanism	Coordinated access	Devices compete
Collision possibility	Generally avoided by the access mechanism	Possible
Waiting	May wait for permission or a turn	May wait, back off, or retransmit
Examples	Polling, token passing, reservation	ALOHA, CSMA/CD, CSMA/CA
Typical application	Token-based networks and scheduled access systems	Shared Ethernet historically; Wi-Fi

ALOHA

ALOHA was developed for packet-radio communication and shared transmission channels. It is an early and simple random-access method.

5.1 Pure ALOHA

In Pure ALOHA, a station can transmit whenever it has a frame ready. A simplified operating sequence is:

1. A station transmits a frame.
2. The receiver checks whether the frame is valid and, in the protocol described here, sends an acknowledgment.
3. The sender waits for the acknowledgment.
4. If the acknowledgment does not arrive within the expected time, the sender assumes the transmission may have failed.
5. The sender waits for a random interval and retransmits the frame, subject to a retry limit.

- If two stations transmit at overlapping times, their frames may collide and become invalid.
- The principal advantage of Pure ALOHA is its simplicity. Its disadvantage is low channel efficiency because frames can overlap whenever their transmission periods conflict.
- The theoretical maximum throughput of Pure ALOHA is approximately 18.4% of the channel's capacity under the standard idealized model.

Slotted ALOHA

- Slotted ALOHA improves Pure ALOHA by dividing time into equal slots. Each slot corresponds to one frame transmission time in the basic model.
- A station may begin transmitting only at the beginning of a slot.
- This arrangement reduces the range of possible collisions because transmissions are aligned to slot boundaries. However, two or more stations may still select the same slot and collide.
- The theoretical maximum throughput of Slotted ALOHA is approximately 36.8%, twice that of Pure ALOHA under the standard model.

Feature	Pure ALOHA	Slotted ALOHA
Transmission start	At any time	At a slot boundary
Synchronization	Not required	Required
Collision probability	Higher	Lower
Maximum theoretical throughput	About 18.4%	About 36.8%

CSMA/CD: Carrier Sense Multiple Access with Collision Detection

CSMA/CD was used in traditional shared, half-duplex Ethernet. The term has three parts:

- Carrier Sense: A station checks whether the medium is busy.
- Multiple Access: Several stations share the medium.
- Collision Detection: A transmitting station detects a collision while transmitting.

Unlike ALOHA, CSMA/CD attempts to reduce collisions by checking the medium before transmission and to limit wasted transmission time by detecting collisions while transmitting.

6.1 How CSMA/CD Works

1. Sense the medium. The station checks whether the channel is idle or busy.
2. Transmit when idle. If the medium is busy, the station waits until it becomes idle.
3. Monitor transmission. The station continues monitoring for a collision while transmitting.
4. Signal a collision. If a collision is detected, the station sends a jamming signal and stops transmitting.
5. Back off and retry. The station waits for a randomized interval determined by the binary exponential backoff procedure before attempting transmission again.

6.2 Why CSMA/CD Is Not Normally Used in Modern Ethernet

Traditional hubs create a shared collision domain. In contrast, an Ethernet switch creates separate link segments, and modern switched Ethernet normally operates in full-duplex mode.

On a full-duplex Ethernet link, a device can transmit and receive simultaneously without competing with other devices for the same collision-prone medium. Consequently, CSMA/CD is not needed on ordinary full-duplex switched Ethernet links.

CSMA/CD is therefore mainly relevant when studying legacy shared Ethernet and the historical development of LAN technologies.

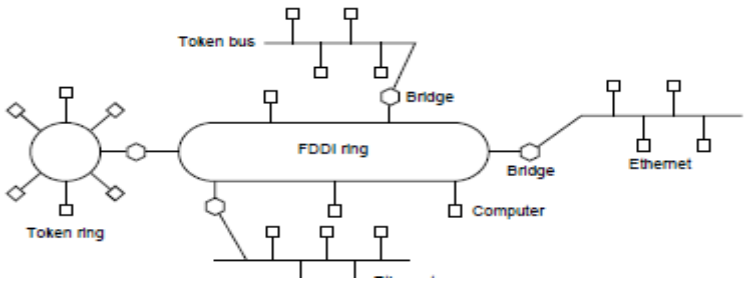
6.3 CSMA/CA in Wi-Fi

Wi-Fi uses CSMA/CA, or Carrier Sense Multiple Access with Collision Avoidance, rather than conventional Ethernet CSMA/CD.

Wireless stations generally cannot reliably detect collisions while transmitting in the same way as traditional wired Ethernet stations. Instead, Wi-Fi uses channel sensing, waiting periods, randomized backoff, and other mechanisms to reduce the likelihood of simultaneous transmissions.

FDDI: Fiber Distributed Data Interface

FDDI is a historical high-speed LAN technology that uses fiber-optic transmission and token-passing access control.



The main characteristics of FDDI include:

- A nominal data rate of 100 Mbps.
- A dual-ring architecture.
- Token passing to coordinate access to the medium.
- Fault-tolerance mechanisms, including ring wrapping under certain failures.
- Support for large network installations.

FDDI is associated with historical fiber-optic campus and backbone networks. Modern Ethernet and other technologies have largely replaced it in most contemporary deployments.

Teaching note: FDDI is useful as a historical example of a LAN technology, but it should not be presented as a currently dominant Ethernet alternative. Its distance and station limits depend on the deployment and standard specifications.

8. Ethernet

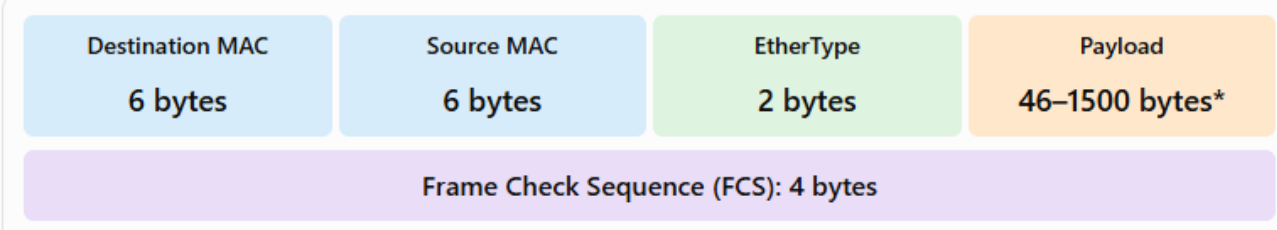
Ethernet is a family of LAN technologies standardized primarily under IEEE 802.3. It defines both Physical Layer characteristics and Data Link Layer framing and MAC behavior.

Ethernet is widely used in office networks, university campuses, data centers, and home networks. Ethernet standards specify matters such as:

- Frame format.
- MAC addressing.
- Media access behavior.
- Physical transmission characteristics.
- Error detection.

8.1 Ethernet Frame Structure

A standard Ethernet II frame has the following structure:



*For a standard untagged Ethernet frame, excluding the preamble and Start Frame Delimiter. VLAN tagging changes the header size and applicable frame-size limits.

The fields are:

- Destination MAC address: Identifies the intended destination interface or group.
- Source MAC address: Identifies the transmitting interface.
- EtherType: Identifies the encapsulated protocol in Ethernet II frames.
- Payload: Contains the encapsulated data. Padding may be added to satisfy the minimum frame size.
- FCS: Contains the error-detection value calculated using CRC.

The Ethernet preamble and Start Frame Delimiter are transmitted before the frame, but are not counted as part of the standard Ethernet frame size.

Ethernet MAC Addressing

A MAC address is a link-layer address used to identify a network interface within the relevant Layer 2 addressing domain.

A conventional Ethernet MAC address is 48 bits (6 bytes) long and is commonly written as 12 hexadecimal digits.

Example: 00:00:0C:12:34:56

- The same address can also be written in Cisco-style dotted hexadecimal notation:
- 0000.0C12.3456

Unicast Address

- A unicast MAC address identifies an individual interface.
- When a frame is addressed to a unicast MAC address, the frame is intended for one network interface.

Broadcast Address

- The Ethernet broadcast address is: FF:FF:FF:FF:FF:FF
- A broadcast frame is intended for all stations in the applicable Layer 2 broadcast domain, subject to network configuration and device behavior.
- Switches normally flood broadcast frames to the other eligible ports in the same VLAN, excluding the port on which the frame arrived.

Multicast Address

- A multicast MAC address identifies a group of interfaces rather than one individual interface or every station in the broadcast domain.
- Multicast allows one transmission to be directed to a group of interested receivers. Switches may use features such as IGMP snooping to limit unnecessary multicast forwarding.
- For IPv4 multicast traffic carried over Ethernet, the destination MAC address uses the range beginning with: 01:00:5E
- The mapping includes only part of the IPv4 multicast address space, so different IP multicast groups can map to the same Ethernet multicast address.
- Important distinction: MAC addresses and IP addresses serve different purposes. MAC addresses support link-layer delivery, while IP addresses support logical addressing and routing across networks.

Data Link Function: Error Detection

Error detection determines whether a frame may have been corrupted during transmission.

Physical transmission can be affected by noise, interference, attenuation, and other signal impairments. These conditions may cause one or more bits to change.

Frame Check Sequence (FCS)

Many Data Link Layer protocols include an error-detection field in the frame or its trailer.

Ethernet uses a four-byte Frame Check Sequence based on a Cyclic Redundancy Check.

The basic process is:

1. The sender calculates a CRC value from the relevant frame contents.
2. The sender places the calculated value in the FCS field.
3. The receiver performs the corresponding error check on the received frame.
4. If the check indicates an error, the frame is treated as corrupted and normally discarded by Ethernet equipment.

CRC and FCS are related but not identical terms. CRC is the calculation method; FCS is the frame field that carries the resulting check value.

Error Detection Versus Error Recovery

Error detection and error recovery are different functions.

- Error detection: Determines whether data may have been corrupted.
- Error recovery: Takes action to obtain correct data, such as retransmitting it.

Ordinary Ethernet does not normally retransmit corrupted frames at the Ethernet MAC layer. Recovery may instead occur at a higher layer. For example, TCP can detect missing or corrupted transport data and arrange retransmission when appropriate.

Data Link Function: Identifying the Encapsulated Data

- The Data Link Layer must determine how the payload of a received frame should be interpreted.
- For example, an Ethernet frame may carry IPv4, ARP, or IPv6 data. The receiver needs to identify the appropriate protocol handler.
- Ethernet II uses a two-byte EtherType field for this purpose.

EtherType	Encapsulated protocol
0x0800	IPv4
0x0806	ARP
0x86DD	IPv6

For example, if the EtherType field contains 0x0800, the receiving Ethernet interface recognizes the payload as IPv4 data and passes it to the IPv4 processing function.

This mechanism supports protocol identification without requiring the receiver to guess the payload type.

Layer 2 Switches

A Layer 2 switch is a network device that forwards Ethernet frames using MAC addresses.

Unlike a traditional hub, which repeats incoming signals to other ports, a switch learns which MAC addresses are reachable through which ports and uses that information to forward frames selectively.

How a Switch Learns MAC Addresses

A switch builds a MAC address table by examining the source MAC address of incoming frames.

A simplified example:

MAC address learned	Switch port
00:11:22:33:44:55	Port 1
00:11:22:33:44:66	Port 2
00:11:22:33:44:77	Port 3

If a frame arrives on Port 1 with source MAC address 00:11:22:33:44:55, the switch records that address as reachable through Port 1.

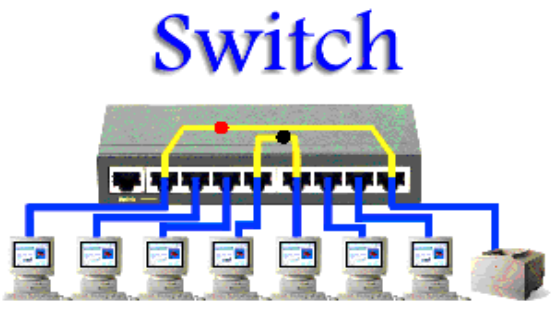
The switch also refreshes learned entries as traffic arrives and eventually ages out inactive dynamic entries according to its configuration.

How a Switch Forwards Frames

A Layer 2 switch generally performs the following operations:

1. Receives a frame.
2. Learns or refreshes the source MAC address and incoming port.
3. Examines the destination MAC address.
4. Checks its MAC address table.
5. Forwards the frame through the appropriate port, filters it when the destination is reachable through the incoming port, or floods it when necessary.

Unknown-unicast and broadcast frames are normally flooded within the relevant VLAN, subject to switch configuration and security features.



Collision Domains and Broadcast Domains

A switch separates collision domains between its Ethernet ports. In modern full-duplex switched Ethernet, collisions do not occur on normal full-duplex links.

However, a Layer 2 switch does not automatically create a separate broadcast domain for every port. Devices in the same VLAN normally share the same broadcast domain. VLANs can be used to divide a switched network into separate broadcast domains.

Ethernet Switching Methods : Two important switching methods are :

- store-and-forward
- and cut-through switching.

Store-and-Forward Switching

- The switch receives and buffers the entire frame before forwarding it.
- The switch can check the FCS before forwarding the frame.

Advantages:

- Can discard frames with invalid FCS values.
- Supports reliable frame-level validation before forwarding.
- Can accommodate differences between incoming and outgoing port speeds by buffering frames.

Disadvantage:

- The switch must receive the complete frame before forwarding begins, increasing latency compared with cut-through switching.

13.2 Cut-Through Switching

The switch begins forwarding a frame after reading enough of the header to identify the destination port.

Advantages:

- Lower forwarding latency for eligible traffic.
- Useful where reducing delay is important.

Disadvantages:

- A corrupted frame may begin being forwarded before the switch has received and checked the complete frame.

- It may propagate bad frames that store-and-forward switching would discard after checking the FCS.

Fragment-Free Switching

Fragment-free switching is a historical compromise between store-and-forward and cut-through switching. It waits for the first 64 bytes before forwarding, aiming to avoid forwarding many collision fragments associated with older shared Ethernet environments.

Feature	Store-and-Forward	Cut-Through	Fragment-Free
Receives entire frame first	Yes	No	No
Can validate the full-frame FCS before forwarding	Yes	Not necessarily	No
Forwarding latency	Generally higher	Generally lower	Intermediate
Main characteristic	Frame validation	Low latency	Historical compromise

Types of Switches

Switches can be classified using several different criteria. These classifications describe different aspects of a switch and are not mutually exclusive.

Based on OSI Layer

Layer 2 switch

- Forwards Ethernet frames using MAC addresses.
- Commonly used to connect end devices within a LAN or VLAN.

Layer 3 switch

- Supports Layer 2 switching and Layer 3 routing.

- Can forward IP packets between VLANs or subnets when configured to do so.

Based on Management

Unmanaged switch

- Requires little or no configuration.
- Typically used in small or simple networks.
- Usually provides limited monitoring and control features.

Managed switch

- Provides configuration and monitoring functions.
- May support VLANs, QoS, link aggregation, access controls, and other enterprise features.

Smart switch

- Offers a more limited set of management functions than many fully managed switches.
- Features vary by manufacturer and model.

Based on Switching Method

- **Store-and-forward**: Receives the entire frame and checks its FCS before forwarding.
- **Cut-through**: Begins forwarding before the entire frame is received.
- **Fragment-free**: Historical approach that waits for the first 64 bytes.

Based on Physical Design

- **Fixed-configuration switch**: Has a fixed set of ports and generally does not support expansion through interchangeable switching modules.
- **Modular switch**: Supports expansion or configuration through modules, depending on the platform. Modular designs are commonly used where scalability and flexible port configurations are important.



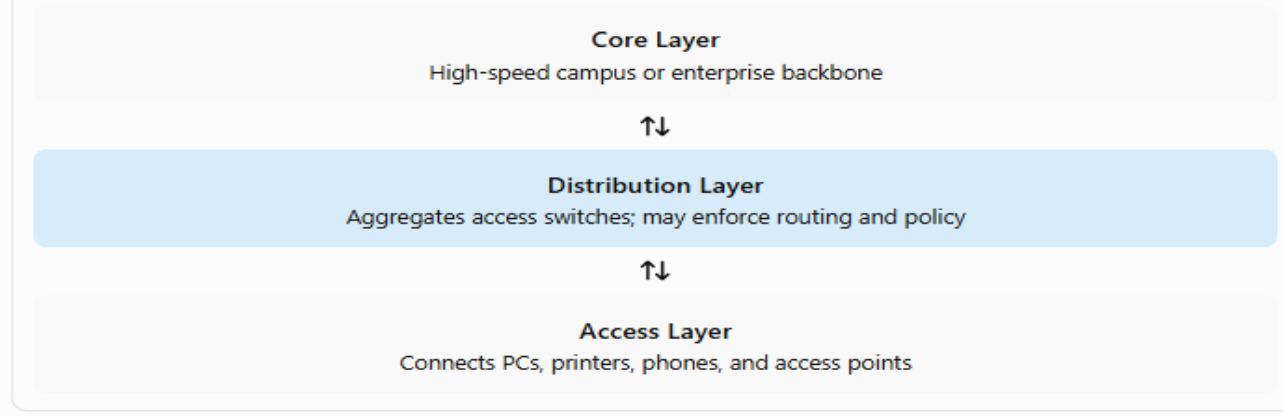
Based on Power Capability

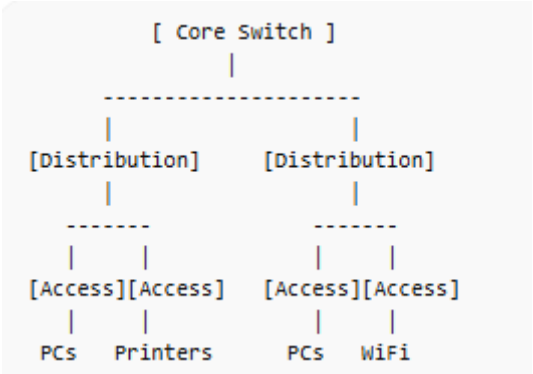
Power over Ethernet (PoE) switch: Can supply electrical power to compatible devices over Ethernet cabling while carrying network data.

Network switches can be classified according to their power capability, particularly whether they can supply electrical power to connected devices through Ethernet cables using Power over Ethernet (PoE).

Switch Type	Description	Typical Applications
Non-PoE Switch	Transfers data only. It does not supply power through Ethernet ports.	PCs, printers, servers
PoE Switch	Transfers data and supplies electrical power through the same Ethernet cable.	IP cameras, VoIP phones, wireless access points
PoE+ Switch	Supports a higher power level than standard PoE.	Advanced IP cameras, dual-band access points
PoE++ Switch	Provides higher power for demanding devices.	Wi-Fi 6/7 access points, PTZ cameras, digital signage

Based on Network Design:





Access switch: Connects end devices such as PCs, printers, IP phones, and wireless access points.

Distribution switch: Aggregates access-layer switches and may perform inter-VLAN routing, access-control enforcement, QoS, and other policy functions.

Core switch: Provides high-speed connectivity between major parts of a campus or enterprise network. Reliability, capacity, and redundancy are important considerations.

A simple analogy is:

- Access: The building entrance that connects users.
- Distribution: The corridor that connects rooms and floors.
- Core: The main highway connecting buildings.

These are network-design roles, not separate OSI layers. A switch may perform more than one role, depending on the network architecture.

Hub Versus Switch

Feature	Hub	Layer 2 Switch
OSI operation	Physical Layer	Data Link Layer
Forwarding behavior	Repeats signals to other ports	Forwards frames based on MAC addresses
Collision domains	Shared collision domain	Separate collision domains per port
Typical operation	Half-duplex shared medium	Commonly full-duplex
Bandwidth use	Shared among connected devices	Dedicated link bandwidth per port, subject to uplink capacity

A hub does not examine MAC addresses. A Layer 2 switch learns and uses MAC addresses to forward frames selectively.